

An Explicit Threshold in Erdős Problem #848

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Mar 24, 2026

Abstract

Let $N \in \mathbb{N}$ and let $A \subseteq \{1, \dots, N\}$ be such that $ab + 1$ is not squarefree for every pair $a, b \in A$. Sawhney proved that, for all sufficiently large N , the truncated residue class 7 modulo 25 is extremal, and the class 18 modulo 25 is also extremal whenever it has the same cardinality [2]. In this note we make the threshold explicit: for every $N \geq 2.64 \times 10^{17}$,

$$|A| \leq |\{n \in [N] : n \equiv 7 \pmod{25}\}| = \left\lfloor \frac{N + 18}{25} \right\rfloor.$$

This bound is attained by $[N] \cap \{n \in \mathbb{Z} : n \equiv 7 \pmod{25}\}$, and also by $[N] \cap \{n \in \mathbb{Z} : n \equiv 18 \pmod{25}\}$ whenever the latter has the same cardinality. Our argument follows Sawhney's mod-25 decomposition and replaces the two asymptotic inputs by explicit estimates for squarefree numbers in arithmetic progressions and for the values of $x^2 + 1$ having a square factor.

Disclaimer: This document is generated in a near-autonomous process by AI, specifically GPT-5.2 Thinking and GPT-5.4 Thinking.¹ It has passed through considerable correctness checks, but mistakes are still possible and caution is advised.

The author thanks Ingo Althofer for improving bounds in some earlier versions of these notes.

1 Introduction

For $N \in \mathbb{N}$, let $[N] := \{1, 2, \dots, N\}$. We study sets $A \subseteq [N]$ with the property that

$$\mu(ab + 1) = 0 \quad (a, b \in A),$$

where μ denotes the Möbius function. Equivalently, every integer of the form $ab + 1$ is not squarefree. A natural construction is to take one of the truncated residue classes

$$[N] \cap \{n \in \mathbb{Z} : n \equiv 7 \pmod{25}\} \quad \text{or} \quad [N] \cap \{n \in \mathbb{Z} : n \equiv 18 \pmod{25}\},$$

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¹<https://chatgpt.com/share/69a33163-d13c-8010-8062-4923ab4b2d34>;
<https://chatgpt.com/share/69a5fd85-bb18-8010-8cf9-0cf91ee1b16a>;
<https://chatgpt.com/share/69a6ca5b-d020-8010-becf-f079cb575cd8>;
<https://chatgpt.com/share/69ab01c2-51b8-8010-9bb0-ce5d02e5fe0e>;
<https://chatgpt.com/share/69ab0f7a-2150-8010-9c82-88b68ce0f7ac>;
<https://chatgpt.com/share/69befe62-dd90-8010-8f74-94d2b215d654>;
<https://chatgpt.com/share/69c03152-2e88-8010-ae98-e3ed3e7ded08>;
<https://chatgpt.com/share/69c042ab-0fc8-8010-998e-cf47aa4cfbe5>;
<https://chatgpt.com/share/69c17376-5b08-8010-9058-f25bc43cf3dd>;
<https://chatgpt.com/share/69c1841a-41ec-8010-b54d-ca4a1ff1cd26>.
Some chats cannot be shared due to persistent errors and are not linked here.

since then $25 \mid ab + 1$ for every pair a, b from that class. The corresponding extremal problem appears as Erdős Problem #848; see Bloom's list [1].

Sawhney [2] proved that, for all sufficiently large N , the truncated residue class 7 modulo 25 is extremal, and the class 18 modulo 25 is also extremal whenever it has the same cardinality. The purpose of this note is to make that threshold explicit. Our main result is the following.

Theorem 1. *Let $N \geq 2.64 \times 10^{17}$, and let $A \subseteq [N]$ satisfy $\mu(ab + 1) = 0$ for all $a, b \in A$. Then*

$$|A| \leq |\{n \in [N] : n \equiv 7 \pmod{25}\}| = \left\lfloor \frac{N + 18}{25} \right\rfloor.$$

Moreover, equality is attained by $[N] \cap \{n \in \mathbb{Z} : n \equiv 7 \pmod{25}\}$, and also by $[N] \cap \{n \in \mathbb{Z} : n \equiv 18 \pmod{25}\}$ whenever the latter has the same cardinality.

The proof follows Sawhney's overall structure. We partition A according to its residue classes modulo 25, write A_7 and A_{18} for the two distinguished classes, and let A^* denote the union of the remaining classes. The final case analysis is the same as in [2]; the main task is to make two auxiliary estimates explicit.

The first estimate concerns squarefree numbers in a fixed arithmetic progression. Here we use a self-contained Möbius-splitting argument, in the spirit of Prachar and Hooley [3, 4], to obtain an explicit replacement for Sawhney's Lemma 2.2. The second estimate bounds the number of x for which $x^2 + 1$ has a square factor; this gives an explicit replacement for Sawhney's Lemma 2.1 by reducing the large-square contribution to a family of negative Pell equations. Combining these bounds with Sawhney's residue-class decomposition yields Theorem 1.

2 An explicit squarefree estimate in an arithmetic progression

For integers $x \geq 1$, $q \geq 1$, and $(a, q) = 1$, write

$$S_2(x; q, a) := \#\{n \leq x : n \equiv a \pmod{q}, \mu^2(n) = 1\}.$$

Also define

$$\rho(q) := \max_{(u, q)=1} \#\{y \pmod{q} : y^2 \equiv u \pmod{q}\}.$$

Lemma 1 (A root-count bound). *For every $q \geq 1$ one has*

$$\rho(q) \leq \lambda q^{1/4}, \quad \lambda := \frac{128}{(8 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13)^{1/4}} < 6.876.$$

Proof. By the Chinese remainder theorem it suffices to consider prime powers. If p is odd, then $y^2 \equiv u \pmod{p^\nu}$ has at most two solutions for every unit u . For powers of 2, the same congruence has exactly one solution modulo 2, at most two solutions modulo 4, and at most four solutions modulo 2^ν for $\nu \geq 3$. Hence, if $q = 2^\nu m$ with m odd, then

$$\rho(q) \leq c_\nu 2^{\omega(m)},$$

where $c_0 = c_1 = 1$, $c_2 = 2$, and $c_\nu = 4$ for $\nu \geq 3$.

If $m = 1$, then

$$\frac{c_\nu}{2^{\nu/4}} \leq \frac{4}{8^{1/4}}$$

for every $\nu \geq 0$, since the left-hand side is maximized at $\nu = 3$. Thus $\rho(q) \leq \lambda q^{1/4}$ in this case.

Now assume $m > 1$ and write $k := \omega(m) \geq 1$. Among odd integers with k distinct prime factors, the minimum is the product of the first k odd primes. A direct check for $1 \leq k \leq 5$ shows that

$$\frac{2^k}{(3 \cdot 5 \cdot 7 \cdots p_k)^{1/4}} \leq \frac{32}{(3 \cdot 5 \cdot 7 \cdot 11 \cdot 13)^{1/4}}.$$

For $k \geq 5$, adjoining the next odd prime multiplies the denominator by at least $17^{1/4} > 2$, while the numerator is multiplied only by 2, so the ratio decreases from that point onward. Therefore

$$2^{\omega(m)} \leq \frac{32}{(3 \cdot 5 \cdot 7 \cdot 11 \cdot 13)^{1/4}} m^{1/4}.$$

Combining this with

$$c_\nu \leq \frac{4}{8^{1/4}} 2^{\nu/4}$$

gives

$$\rho(q) \leq \frac{4}{8^{1/4}} 2^{\nu/4} \cdot \frac{32}{(3 \cdot 5 \cdot 7 \cdot 11 \cdot 13)^{1/4}} m^{1/4} = \lambda q^{1/4}.$$

□

Proposition 1 (Explicit squarefree count in a progression). *Let $x \geq 1$, let $q \geq 1$ with $q \leq x$, let $(a, q) = 1$, and let $D \geq 1$. Then*

$$\left| S_2(x; q, a) - \frac{x}{q} \prod_{p \nmid q} \left(1 - \frac{1}{p^2} \right) \right| \leq D + \frac{x}{qD} + \lambda q^{1/4} \left(\frac{2x}{qD} + \frac{x}{D^2} \right).$$

Proof. The identity

$$\mu^2(n) = \sum_{d^2 | n} \mu(d)$$

gives

$$S_2(x; q, a) = \sum_{d \geq 1} \mu(d) A_d, \quad A_d := \#\{m \leq x/d^2 : md^2 \equiv a \pmod{q}\}.$$

Split the sum at D .

If $(d, q) > 1$, then $A_d = 0$ because $(a, q) = 1$. If $(d, q) = 1$, then $md^2 \equiv a \pmod{q}$ picks out one residue class modulo q , so

$$\left| A_d - \frac{x}{qd^2} \right| \leq 1.$$

Therefore

$$\left| \sum_{d \leq D} \mu(d) A_d - \frac{x}{q} \sum_{\substack{d \leq D \\ (d, q) = 1}} \frac{\mu(d)}{d^2} \right| \leq D.$$

Also,

$$\left| \frac{x}{q} \sum_{\substack{d \leq D \\ (d, q) = 1}} \frac{\mu(d)}{d^2} - \frac{x}{q} \prod_{p \nmid q} \left(1 - \frac{1}{p^2} \right) \right| \leq \frac{x}{q} \sum_{d > D} \frac{1}{d^2} < \frac{x}{qD}.$$

Hence

$$\left| \sum_{d \leq D} \mu(d) A_d - \frac{x}{q} \prod_{p \nmid q} \left(1 - \frac{1}{p^2}\right) \right| \leq D + \frac{x}{qD}.$$

For the tail,

$$\left| \sum_{d > D} \mu(d) A_d \right| \leq \sum_{d > D} A_d.$$

Write $n = md^2$. Then $m \leq x/D^2$, $(m, q) = 1$, and for each fixed m the congruence $d^2 \equiv a\bar{m} \pmod{q}$, where $\bar{m}$ denotes the inverse of m modulo q , has at most $\rho(q)$ residue classes modulo q . Thus

$$\begin{aligned} \sum_{d > D} A_d &\leq \sum_{m \leq x/D^2} \#\left\{d \leq \sqrt{x/m} : d^2 \equiv a\bar{m} \pmod{q}\right\} \\ &\leq \rho(q) \sum_{m \leq x/D^2} \left(\frac{\sqrt{x/m}}{q} + 1\right) \\ &\leq \rho(q) \left(\frac{2x}{qD} + \frac{x}{D^2}\right). \end{aligned}$$

By Lemma 1,

$$\left| \sum_{d > D} \mu(d) A_d \right| \leq \lambda q^{1/4} \left(\frac{2x}{qD} + \frac{x}{D^2}\right).$$

Combining the bounds gives the result. $\square$

Corollary 1 (Explicit replacement for Sawhney's Lemma 2.2). *Let $q_0 \in \{25, 100\}$, let $b \in \mathbb{N}$ with $b \leq N$, and let $t \pmod{q_0}$ satisfy*

$$p^2 \nmid (bt + 1) \quad \text{for every prime } p \mid q_0.$$

Then, for every $N \geq 2q_0$,

$$\#\{a \leq N : a \equiv t \pmod{q_0}, \mu(ab + 1) = 0\} \leq \frac{N}{q_0} \left(1 - \prod_{(p, q_0 b) = 1} \left(1 - \frac{1}{p^2}\right)\right) + \kappa_{q_0} N^{3/4},$$

where one may take

$$\kappa_{25} = 4.7, \quad \kappa_{100} = 5.3.$$

Proof. Choose $a_0 \in [1, q_0]$ with $a_0 \equiv t \pmod{q_0}$ and write

$$a = a_0 + q_0 n, \quad 0 \leq n < X, \quad X := \#\{a \leq N : a \equiv t \pmod{q_0}\}.$$

Set

$$d := q_0 b, \quad c := ba_0 + 1, \quad g := \gcd(c, d).$$

Then $ab + 1 = dn + c$.

If $p \mid g$, then $p \nmid b$ because $c \equiv 1 \pmod{p}$ for every prime $p \mid b$. Hence $p \mid q_0$. Since q_0 is a perfect square, $p^2 \mid d$. By hypothesis $p^2 \nmid c$, so p occurs to exact exponent 1 in g . Thus g is squarefree. Moreover, if $p \mid g$, then $p \mid q_0$ and $p^2 \mid q_0$, so $p \mid d/g$. Writing

$$d = gd_1, \quad c = gc_1,$$

we therefore have $(c_1, d_1) = 1$, every prime divisor of g also divides d_1 , and hence

$$d_1 n + c_1 \equiv c_1 \not\equiv 0 \pmod{p} \quad (p \mid g)$$

for every n . Therefore

$$\mu^2(dn + c) = \mu^2(g(d_1 n + c_1)) = \mu^2(d_1 n + c_1)$$

for every n . The prime divisors of d_1 are exactly the prime divisors of $d = q_0 b$.

The values $d_1 n + c_1$ with $0 \leq n < X$ form an arithmetic progression of length X in the residue class $c_1 \pmod{d_1}$. Let a_1 be the least positive residue congruent to c_1 modulo d_1 . Since

$$c_1 = \frac{ba_0 + 1}{g} \leq \frac{bq_0 + 1}{g} = d_1 + \frac{1}{g} \leq d_1 + 1,$$

we have either $c_1 = a_1$ or $c_1 = d_1 + 1$ and $a_1 = 1$. Thus the set $\{d_1 n + c_1 : 0 \leq n < X\}$ differs from the set of positive integers $m \leq x := d_1(X - 1) + c_1$ with $m \equiv a_1 \pmod{d_1}$ by at most the single initial value 1. Since $N \geq 2q_0$, every residue class modulo q_0 occurs at least twice in $[1, N]$, so $X \geq 2$ and hence $x \geq d_1 + 1 > d_1$. Also $c_1 \leq d_1 + 1$, so

$$\frac{x}{d_1} = X - 1 + \frac{c_1}{d_1} \leq X + \frac{1}{d_1} \quad \text{and} \quad x \leq d_1 X + 1.$$

Apply Proposition 1 with modulus d_1 , residue a_1 , and

$$D := \left\lceil \alpha_{q_0} N^{3/4} \right\rceil, \quad \alpha_{25} := \frac{25}{8}, \quad \alpha_{100} := \frac{7}{2}.$$

Since $x/d_1 = X - 1 + c_1/d_1 \geq X - 1$, this gives

$$\begin{aligned} \#\{0 \leq n < X : \mu^2(d_1 n + c_1) = 1\} &\geq \frac{x}{d_1} \prod_{(p, q_0 b)=1} \left(1 - \frac{1}{p^2}\right) - 1 - D - \frac{x}{d_1 D} \\ &\quad - \lambda d_1^{1/4} \left(\frac{2x}{d_1 D} + \frac{x}{D^2} \right) \\ &\geq (X - 1) \prod_{(p, q_0 b)=1} \left(1 - \frac{1}{p^2}\right) - 1 - D - \frac{X}{D} - \frac{1}{d_1 D} \\ &\quad - 2\lambda d_1^{1/4} \frac{X}{D} - \frac{2\lambda}{d_1^{3/4} D} - \lambda d_1^{5/4} \frac{X}{D^2} - \frac{\lambda d_1^{1/4}}{D^2}. \end{aligned}$$

Hence

$$\begin{aligned} \#\{a \leq N : a \equiv t \pmod{q_0}, \mu(ab + 1) = 0\} &\leq X \left(1 - \prod_{(p, q_0 b)=1} \left(1 - \frac{1}{p^2}\right)\right) + 2 + D + \frac{X}{D} + \frac{1}{d_1 D} \\ &\quad + 2\lambda d_1^{1/4} \frac{X}{D} + \frac{2\lambda}{d_1^{3/4} D} + \lambda d_1^{5/4} \frac{X}{D^2} + \frac{\lambda d_1^{1/4}}{D^2}. \end{aligned}$$

Now g divides the squarefree kernel of q_0 , so $q_0/g \geq \sqrt{q_0}$; since $d_1 = q_0 b/g$ and $b \geq 1$, it follows that

$$\sqrt{q_0} \leq d_1 \leq q_0 N, \quad X \leq \frac{N}{q_0} + 1.$$

Also,

$$0 \leq 1 - \prod_{(p,q_0b)=1} \left(1 - \frac{1}{p^2}\right) \leq 1 - \prod_{p \neq 5} \left(1 - \frac{1}{p^2}\right) < 0.367,$$

since $q_0 \in \{25, 100\}$. Hence replacing X by N/q_0 in the main term costs at most 0.367.

For each fixed $q_0 \in \{25, 100\}$, divide the remaining error terms by $N^{3/4}$. Using

$$D \leq \alpha_{q_0} N^{3/4} + 1, \quad \frac{1}{D} \leq \frac{1}{\alpha_{q_0} N^{3/4}}, \quad \frac{1}{D^2} \leq \frac{1}{\alpha_{q_0}^2 N^{3/2}},$$

and also

$$\sqrt{q_0} \leq d_1 \leq q_0 N, \quad X \leq \frac{N}{q_0} + 1,$$

we obtain the upper bound

$$\begin{aligned} E_{q_0}(N) := & \frac{2.367}{N^{3/4}} + \alpha_{q_0} + \frac{1}{N^{3/4}} + \frac{N/q_0 + 1}{\alpha_{q_0} N^{3/2}} + \frac{1}{\alpha_{q_0} \sqrt{q_0} N^{3/2}} \\ & + \frac{2\lambda q_0^{1/4} (N/q_0 + 1)}{\alpha_{q_0} N^{5/4}} + \frac{2\lambda}{\alpha_{q_0} q_0^{3/8} N^{3/2}} + \frac{\lambda q_0^{5/4} (N/q_0 + 1)}{\alpha_{q_0}^2 N} + \frac{\lambda q_0^{1/4}}{\alpha_{q_0}^2 N^2}. \end{aligned}$$

Each nonconstant term on the right is decreasing in N , so $E_{q_0}(N)$ is decreasing. At $N = 2.64 \times 10^{17}$ one finds

$$E_{25}(N) < 4.700, \quad E_{100}(N) < 5.275.$$

Therefore the stated choices

$$\kappa_{25} = 4.7, \quad \kappa_{100} = 5.3$$

are admissible. □

3 An explicit bound for the $x^2 + 1$ progression

Define

$$C_{\text{quad}} := 1 - \prod_{\substack{p \equiv 1 \pmod{4} \\ p \geq 13}} \left(1 - \frac{2}{p^2}\right).$$

Lemma 2 (A numerical bound for C_{quad}). *One has*

$$C_{\text{quad}} < 0.02734669.$$

Proof. A direct computation up to 10^7 gives

$$\prod_{\substack{13 \leq p \leq 10^7 \\ p \equiv 1 \pmod{4}}} \left(1 - \frac{2}{p^2}\right) = 0.9726535106 \dots$$

For the tail, using $\prod_i (1 - u_i) \geq 1 - \sum_i u_i$ for $0 \leq u_i \leq 1$ and $\sum_{n > y} n^{-2} < 1/(y - 1)$,

$$\prod_{\substack{p > 10^7 \\ p \equiv 1 \pmod{4}}} \left(1 - \frac{2}{p^2}\right) \geq 1 - \sum_{p > 10^7} \frac{2}{p^2} > 1 - \frac{2}{10^7 - 1}.$$

Combining these bounds yields the claim. □

Proposition 2 (Explicit replacement for Sawhney's Lemma 2.1). *Let $q \in \{25, 50\}$, let $t \pmod{q}$, and assume $t \not\equiv 7, 18 \pmod{25}$. Write*

$$X := \#\{x \leq N : x \equiv t \pmod{q}\}.$$

Then for every integer $R \geq 2$,

$$\begin{aligned} \#\{x \leq N : x \equiv t \pmod{q}, \mu(x^2 + 1) = 0\} &\leq X C_{\text{quad}} + R(1 + \log R) \\ &\quad + \frac{2X(\log R + 2)}{R} + \frac{N^2 + 1}{R^2} \left(1 + \log_{2+\sqrt{3}} N\right). \end{aligned}$$

Proof. Write $x = qn + t$ with $0 \leq n < X$ and let

$$Q := \#\{0 \leq n < X : \mu^2((qn + t)^2 + 1) = 1\}.$$

Then

$$Q = \sum_{r \geq 1} \mu(r) A_r, \quad A_r := \#\{0 \leq n < X : r^2 \mid (qn + t)^2 + 1\}.$$

If r has a prime factor $p \mid q$, then $A_r = 0$, since $p = 2$ or 5 and $x^2 + 1$ is never divisible by 4 or by 25 on the progression under consideration. If r has a prime factor $p \equiv 3 \pmod{4}$, then also $A_r = 0$ because -1 is not a quadratic residue modulo p . Hence only squarefree r composed of primes $p \equiv 1 \pmod{4}$, $p \geq 13$, contribute. For such r , the congruence $u^2 \equiv -1 \pmod{r^2}$ has exactly $2^{\omega(r)}$ solutions modulo r^2 . Hence

$$\left| A_r - \frac{X 2^{\omega(r)}}{r^2} \right| \leq 2^{\omega(r)}.$$

Set

$$T_R := \sum_{r > R} A_r.$$

Then, since $|\mu(r)| \leq 1$,

$$Q \geq X \sum_{r \leq R} \mu(r) \frac{2^{\omega(r)}}{r^2} - \sum_{r \leq R} 2^{\omega(r)} - T_R.$$

Since $2^{\omega(r)} \leq d(r)$ and

$$\sum_{r \leq R} d(r) = \sum_{ab \leq R} 1 \leq R \sum_{a \leq R} \frac{1}{a} \leq R(1 + \log R),$$

we obtain

$$\sum_{r \leq R} 2^{\omega(r)} \leq R(1 + \log R).$$

Also, with the same notation and partial summation,

$$\sum_{r > R} \frac{2^{\omega(r)}}{r^2} \leq \sum_{r > R} \frac{d(r)}{r^2} \leq \frac{2(\log R + 2)}{R}.$$

Therefore

$$\sum_{r \leq R} \mu(r) \frac{2^{\omega(r)}}{r^2} \geq \prod_{\substack{p \equiv 1 \pmod{4} \\ p \geq 13}} \left(1 - \frac{2}{p^2}\right) - \frac{2(\log R + 2)}{R}.$$

Hence

$$Q \geq X \prod_{\substack{p \equiv 1 \pmod{4} \\ p \geq 13}} \left(1 - \frac{2}{p^2}\right) - R(1 + \log R) - \frac{2X(\log R + 2)}{R} - T_R.$$

Therefore

$$\#\{x \leq N : x \equiv t \pmod{q}, \mu(x^2 + 1) = 0\} \leq X C_{\text{quad}} + R(1 + \log R) + \frac{2X(\log R + 2)}{R} + T_R.$$

It remains to bound T_R . Each pair counted by A_r corresponds to an integer $x \leq N$ with $x \equiv t \pmod{q}$ and $r^2 \mid x^2 + 1$. Writing

$$x^2 + 1 = dr^2,$$

we have

$$d \leq \frac{N^2 + 1}{R^2}, \quad x^2 - dr^2 = -1.$$

For each fixed positive integer d , the positive solutions of $x^2 - du^2 = -1$ grow at least geometrically. Indeed, if

$$x_2 + u_2\sqrt{d} > x_1 + u_1\sqrt{d} > 1$$

are two positive solutions, then their quotient is a totally positive unit of norm 1, hence has the shape $v + w\sqrt{d}$ with integers v, w , $w \geq 1$, and $v^2 - dw^2 = 1$. Since $x_1^2 - du_1^2 = -1$, one has $x_1 < u_1\sqrt{d}$. Expanding

$$x_2 + u_2\sqrt{d} = (x_1 + u_1\sqrt{d})(v + w\sqrt{d})$$

therefore gives

$$x_2 = x_1v + du_1w > x_1(v + \sqrt{d}w).$$

Now $v + \sqrt{d}w \geq 2 + \sqrt{3}$. Indeed, if $v = 2$, then $dw^2 = 3$, so necessarily $v + \sqrt{d}w = 2 + \sqrt{3}$; if $v \geq 3$, then $v + \sqrt{d}w \geq 3 + \sqrt{2} > 2 + \sqrt{3}$. Thus

$$x_2 > (2 + \sqrt{3})x_1.$$

Therefore, for each fixed d , the number of positive solutions (x, u) with $x \leq N$ is at most $1 + \log_{2+\sqrt{3}} N$. Summing over $d \leq (N^2 + 1)/R^2$ gives

$$T_R \leq \frac{N^2 + 1}{R^2} \left(1 + \log_{2+\sqrt{3}} N\right).$$

Substituting this into the previous estimate finishes the proof. $\square$

Corollary 2 (Numerical bound for A^*). *Let*

$$\begin{aligned} A_7 &:= \{a \in A : a \equiv 7 \pmod{25}\}, \\ A_{18} &:= \{a \in A : a \equiv 18 \pmod{25}\}, \\ A^* &:= A \setminus (A_7 \cup A_{18}). \end{aligned}$$

For every $N \geq 2.64 \times 10^{17}$ one has

(a)

$$\frac{|A^*|}{N} \leq \frac{23}{25} C_{\text{quad}} + 0.001973;$$

(b) if A^* consists only of odd elements, then

$$\frac{|A^*|}{N} \leq \frac{23}{50}C_{\text{quad}} + 0.001973.$$

Proof. Apply Proposition 2 to each of the 23 admissible residue classes modulo 25 in part (a), and to each of the 23 admissible odd residue classes modulo 50 in part (b). In either case, if $q \in \{25, 50\}$ denotes the modulus in question and $X_t := \#\{x \leq N : x \equiv t \pmod{q}\}$, then

$$\sum_t X_t \leq 23 \left(\frac{N}{q} + 1 \right),$$

where the sum runs over the corresponding 23 admissible residue classes. Thus the total main-term contribution is at most

$$\frac{23}{q}NC_{\text{quad}} + 23C_{\text{quad}}.$$

Take

$$R := \left\lceil \frac{259}{200}N^{2/3} \right\rceil.$$

Since $R \leq \frac{259}{200}N^{2/3} + 1$, the quantity

$$\frac{R(1 + \log R)}{N} + \frac{2(N/q + 1)(\log R + 2)}{NR} + \frac{N^2 + 1}{NR^2} \left(1 + \log_{2+\sqrt{3}} N \right)$$

is bounded above by

$$\begin{aligned} & \left(\frac{259}{200}N^{-1/3} + N^{-1} \right) \left(1 + \log \left(\frac{259}{200}N^{2/3} + 1 \right) \right) \\ & + \frac{400(N/q + 1)}{259N^{5/3}} \left(\log \left(\frac{259}{200}N^{2/3} + 1 \right) + 2 \right) \\ & + \frac{40000(N^2 + 1)}{67081N^{7/3}} \left(1 + \log_{2+\sqrt{3}} N \right). \end{aligned}$$

For $q \in \{25, 50\}$, a direct check shows that the right-hand side is decreasing for $N \geq 2.64 \times 10^{17}$. At $N = 2.64 \times 10^{17}$ it is less than 8.577×10^{-5} . Multiplying by 23 gives a density error less than 1.973×10^{-3} . Also,

$$\frac{23C_{\text{quad}}}{N} < 3 \times 10^{-18} \quad (N \geq 2.64 \times 10^{17}),$$

so the extra contribution from the preceding main-term estimate is negligible on this scale. Rounding upward to 0.001973 proves both claims. $\square$

4 Numerical constants

Define

$$C_{2,5} := 1 - \prod_{p \neq 2,5} \left(1 - \frac{1}{p^2} \right), \quad C_5 := 1 - \prod_{p \neq 5} \left(1 - \frac{1}{p^2} \right).$$

Since $\prod_p (1 - p^{-2}) = 1/\zeta(2) = 6/\pi^2$,

$$C_{2,5} = 1 - \frac{6/\pi^2}{(1 - 1/2^2)(1 - 1/5^2)} = 0.1556568029 \dots, \tag{1}$$

$$C_5 = 1 - \frac{6/\pi^2}{(1 - 1/5^2)} = 1 - \frac{25}{4\pi^2} = 0.3667426022 \dots \tag{2}$$

5 Proof of the main theorem

Proof of Theorem 1. The residue class $7 \pmod{25}$ satisfies $ab + 1 \equiv 0 \pmod{25}$ for all a, b in the class, so it meets the constraint and has size $\lfloor (N + 18)/25 \rfloor$. The same is true for the residue class $18 \pmod{25}$, whose cardinality is $\lfloor (N + 7)/25 \rfloor$. It therefore remains to prove the stated upper bound. As in Sawhney's argument, define

$$\begin{aligned} A_7 &:= \{a \in A : a \equiv 7 \pmod{25}\}, \\ A_{18} &:= \{a \in A : a \equiv 18 \pmod{25}\}, \\ A^* &:= A \setminus (A_7 \cup A_{18}). \end{aligned}$$

If $A \subseteq A_7$, then the result is immediate. If $A \subseteq A_{18}$, then

$$|A| \leq |\{n \in [N] : n \equiv 18 \pmod{25}\}| = \left\lfloor \frac{N + 7}{25} \right\rfloor \leq \left\lfloor \frac{N + 18}{25} \right\rfloor.$$

So it suffices to show that whenever A is not contained in A_7 or A_{18} , one has

$$\frac{|A|}{N} < \frac{1}{25}.$$

We consider three cases.

Case 1: A^* contains an even element. Pick an even $b \in A^*$. For every $a \in A_7 \cup A_{18}$, the integer $ab + 1$ is odd and is not divisible by 25. Hence Corollary 1 with $q_0 = 25$ gives

$$\frac{|A_7 \cup A_{18}|}{N} \leq \frac{2}{25} C_{2,5} + 9.4N^{-1/4}.$$

By Corollary 2(a),

$$\frac{|A^*|}{N} \leq \frac{23}{25} C_{\text{quad}} + 0.001973.$$

Therefore

$$\frac{|A|}{N} \leq \frac{23}{25} C_{\text{quad}} + \frac{2}{25} C_{2,5} + 0.001973 + 9.4N^{-1/4}.$$

Using Lemma 2, (1), and $N \geq 2.64 \times 10^{17}$, we obtain

$$\frac{|A|}{N} < 0.0376115 + 0.001973 + 0.000415 < 0.04.$$

Case 2: $A^* \neq \emptyset$ and A^* consists only of odd elements. By Corollary 2(b),

$$\frac{|A^*|}{N} \leq \frac{23}{50} C_{\text{quad}} + 0.001973.$$

Subcase 2a: $A_7 \cup A_{18}$ contains no even elements. Then

$$A_7 \subseteq \{7, 57\} \pmod{100}, \quad A_{18} \subseteq \{43, 93\} \pmod{100}.$$

Fix $b \in A^*$. Since b is odd, among $\{7, 57\}$ exactly one residue class gives $4 \mid (bt + 1)$, and among $\{43, 93\}$ exactly one residue class gives $4 \mid (bt + 1)$. Also $25 \nmid (bt + 1)$ on all four residue classes, because $t \equiv 7$ or $18 \pmod{25}$ whereas $b \in A^*$ implies $b \not\equiv 7, 18 \pmod{25}$. These two classes

contribute at most $2(N/100 + 1)$ elements. On the remaining two residue classes, Corollary 1 with $q_0 = 100$ applies and yields

$$\frac{|A_7 \cup A_{18}|}{N} \leq \frac{1}{50} + \frac{1}{50}C_{2,5} + 10.6N^{-1/4} + \frac{2}{N}.$$

Hence

$$\frac{|A|}{N} \leq \frac{23}{50}C_{\text{quad}} + \frac{1}{50} + \frac{1}{50}C_{2,5} + 0.001973 + 10.6N^{-1/4} + \frac{2}{N}.$$

Using Lemma 2, (1), and $N \geq 2.64 \times 10^{17}$, we obtain

$$\frac{|A|}{N} < 0.0356927 + 0.001973 + 0.000468 + 10^{-17} < 0.04.$$

Subcase 2b: at least one of A_7 or A_{18} contains an even element. Without loss of generality, assume A_7 contains an even element and pick $b' \in A_7$ even. Pick any $b \in A^*$. Since $b \not\equiv 7 \pmod{25}$, we have $25 \nmid (7b + 1)$, so Corollary 1 applies to A_7 with the odd witness b and gives

$$\frac{|A_7|}{N} \leq \frac{1}{25}C_5 + 4.7N^{-1/4},$$

while $b' \equiv 7 \pmod{25}$ implies $25 \nmid (18b' + 1)$, so applying it to A_{18} with the even witness b' gives

$$\frac{|A_{18}|}{N} \leq \frac{1}{25}C_{2,5} + 4.7N^{-1/4}.$$

Therefore

$$\frac{|A|}{N} \leq \frac{23}{50}C_{\text{quad}} + \frac{1}{25}C_5 + \frac{1}{25}C_{2,5} + 0.001973 + 9.4N^{-1/4}.$$

Using Lemma 2, (1)–(2), and $N \geq 2.64 \times 10^{17}$, we obtain

$$\frac{|A|}{N} < 0.0334755 + 0.001973 + 0.000415 < 0.04.$$

Case 3: $A^* = \emptyset$ and both A_7 and A_{18} are nonempty. Pick $b \in A_7$ and $b' \in A_{18}$. Since $b \equiv 7 \pmod{25}$ and $b' \equiv 18 \pmod{25}$, we have $25 \nmid (18b + 1)$ and $25 \nmid (7b' + 1)$. Thus Corollary 1 with $q_0 = 25$ applies to A_7 (using b') and to A_{18} (using b), yielding

$$\frac{|A|}{N} \leq \frac{2}{25}C_5 + 9.4N^{-1/4}.$$

Using (2) and $N \geq 2.64 \times 10^{17}$, we obtain

$$\frac{|A|}{N} < 0.0293395 + 0.000415 < 0.04.$$

All cases give $|A|/N < 1/25$, which finishes the proof. $\square$

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